

University of Asia Pacific (UAP)
Department of Computer Science and Engineering (CSE)
BSc in CSE Program

Course Outline – Object Oriented Programming I

Part A – Introduction

1. **Course No. / Course Code:** CSE 201
2. **Course Title:** Object-Oriented Programming I
3. **Course Type:** Core Course
4. **Level/Term and Section:** 3rd Semester (2nd Year/1st Semester)
5. **Academic Session:** Spring 26
6. **Course Instructor:** Nuzhat Tabassum Progga, Md. Abu Sayem
7. **Prerequisite (If any):** CSE 103
8. **Credit Value:** 3.0
9. **Contact Hours:** 3.0
10. **Total Marks:** 100
11. **Course Objectives and Course Summary:**

The objectives of this course are to:

1. Teach OOP principles and features and how to apply them in a specific object-oriented language.
2. Demonstrate how to properly utilize the Exception Handling mechanism and write multithreaded applications.
3. Show how to use the advanced library ("Collections & Generics" or "STL & Templates").

This course will cover the fundamental aspects of object-oriented programming. Students will learn how to apply object-oriented programming principles effectively using a specific object-oriented language. The course begins with the language's basic syntax and gradually introduces its object-oriented features. Topics include core OOP principles, data types, variables, variable scope and lifetime, operators, classes and objects, inheritance, abstraction, exception

handling, multithreading, file handling, tokenization, networking, input/output streams, and advanced library components such as collections or templates.

12. Course Learning Outcomes: at the end of the Course, the Student will be able to –

CLO 1	Explain the basics of Object-Oriented Programming Features.
CLO 2	Develop applications using programming language basics.
CLO 3	Develop Object-Oriented solutions for programming problems.
CLO 4	Analyze and Debug program.
CLO 5	Create applications using the language-specific library.

13. Mapping / Alignment of CLOs with Program Learning Outcomes (PLO) (Optional):

CLO No.	Corresponding PLOs (Appendix-1)	Bloom's taxonomy domain/level (Appendix-2)	Delivery methods and activities	Assessment Tools
CLO1	a	1/Understand	Lecture, multimedia	Viva, Presentation, Written exam
CLO2	c	1/Apply	Lecture, Problem Analysis, and Implementation.	Quiz, Written exam
CLO3	c	1/Apply	Lecture, Problem Analysis, and Implementation.	Quiz, Written exam
CLO4	b	1/Analyze	Lecture, Problem Analysis, and Implementation.	Quiz, Written Exam
CLO5	c	1/Apply	Lecture, Problem Analysis, and Implementation.	Problem Solving, Written Exam

Part B – Content of the Course

14. Course Content:

15. Alignment of topics of the courses with CLOs:

SL. No	Topics / Content	Course Learning Outcome (CLO)
1	Language Basic	CLO1, CLO2, CLO4

2	OOP Basics	CLO1, CLO3, CLO4
3	OOP Principles: Inheritance, Encapsulation, Polymorphism, Abstraction	CLO1, CLO3, CLO4
4.	Language Specific library (Exception, Thread, IO, Collections	CLO1, CLO4, CLO5

16. Class Schedule/Lesson Plan/Weekly plan:

Topics	Specific Outcome(s)	Time Frame	Suggested Activities	Teaching Strategy(s)	Alignment with CLO
Introduction to Java and Object Oriented Programming principles, Data types, Variables, Scoping and lifetime of variable	• Understand basic syntax and structure of Java programs • Identify primitive data types and variables • Explain variable scope and lifetime	Week 1	Code walkthroughs, introductory coding exercises, Java setup & environment tasks	Lecture, multimedia	CLO1, CLO2
Arrays: Single and Multidimensional Operators: Arithmetic, The Bitwise Operators, Boolean Logical Operators, Relational Operators, Precedence Control Statements	• Use different types of operators in Java • Implement control flow using if, switch, loops • Declare and manipulate arrays	Week 2		Lecture, multimedia	CLO1, CLO2
Introduction to classes and objects, Constructors and methods. Reference type as parameter and return type. Package, Static keyword, String and String Tokenizer. Wrapper Class	• Create classes and objects • Define and use constructors and methods • Work with Java packages and static members • Use wrapper classes and String utilities	Week 3-5	CT on Java Basics	Lecture, Problem Solving	CLO1, CLO3, CLO4

OOP Features: Inheritance, Encapsulation, Method overloading, Polymorphism, method overriding, Final keyword	• Implement inheritance and encapsulation • Apply polymorphism • Explain use of the final keyword	Week 6-7	CT on Class and Object	Lecture, Problem Solving, Group discussion	CLO1, CLO3, CLO4
MID-TERM EXAMINATION					
Abstraction: abstract class, Interface	• Differentiate between abstract classes and interfaces • Apply abstraction in program design	Week 8	CT on OOP Features. Assignment on Abstraction	Lecture, multimedia	CLO1, CLO3, CLO4
Exception Handling	• Handle runtime errors using try-catch-finally • Define custom exceptions	Week 9		Lecture, multimedia	CLO1, CLO4, CLO5
Nested class, Threading	• Implement inner/nested classes • Create and manage threads using Thread and Runnable	Week 10-11	CT on Exception and Thread	Lecture, multimedia	CLO1, CLO4, CLO5
Input/Output and Serialization	• Read/write files in Java • Implement object serialization & deserialization	Week 12		Lecture, multimedia	CLO1, CLO4, CLO5
Collections and Generics	• Use List, Set, Map interfaces and their implementations • Apply generics for type safety	Week 13-14	IO	Lecture, multimedia	CLO1, CLO4, CLO5
FINAL EXAMINATION					

17. Teaching-Learning Strategies: Interactive Lectures, Google Classroom

18. Assessment Techniques of each topic of the course:

SL. No	Topics / Content	Assessment Techniques
1	Language Basic	Class Test, Written Exam
2	OOP Basics	Class Test, Written Exam
3	OOP Principles: Inheritance, Encapsulation, Polymorphism, Abstraction	QA Session, Assignment, Class Test, Written Exam
4.	Language Specific library (Exception, Thread, IO, Collections	Class Test, Written Exam

Part C – Assessment and Evaluation

19. Assessment Strategy

Class Tests: Altogether 5 class tests may be taken during the semester, 3 class tests will be taken for midterm and 2 class tests will be taken for final term. 3 out of 5 class tests will be considered. CT1, best of CT2 & CT3, and best of CT4 & CT5 will be considered. No makeup class tests will be taken. Students are strongly recommended not to miss any class tests.

Assignment: Assignment (Written and/or presentation.) will be given throughout the semester. Late submission will result in a 50% deduction in score. Students will have to do the presentation on the given topic as an assignment.

CIE- Continuous Internal Evaluation (30 Marks)

Bloom's Category Marks (out of 30)	Tests (20)	Assignments (10)
Remember		
Understand		
Apply	20	10
Analyze		

Evaluate		
Create		

SMEB- Semester Mid & End Examination (70 Marks)

Bloom's Category	Test
Remember	14
Understand	
Apply	42
Analyze	14
Evaluate	
Create	

Weighting COs with Assessment methods:

Assessment Type		Marks Distribut ion (%)	CLO1	CLO2	CLO3	CLO4	CLO5
			PLO- a	PLO-c	PLO- c	PLO-b	PLOc
Final Exam (50%)	Written Exam	50	10	10	10	10	10
Mid Term (20%)	Written Exam	20	4	4	8	4	
Assessment (30%)	Assignment/Pres entation/Viva	10			10		
	Quiz	20		5	10		5
Total		100%	14	19	38	14	15

20. Evaluation Policy

Grades will be calculated as per the university grading structure and individual students will be evaluated based on the following criteria with respective weights.

1. Assessment 30%
2. Term Examination 50%
3. Mid-Term Examination 20%

UAP Grading Policy

Numeric Grade	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Part D – Learning Resources

21. Text Book

1. Herbert Schildt and Danny Coward, Java the Complete Reference, 12th edition (2021), McGraw Hill

Reference Books & Materials

1. Paul Deitel and Harvey M. Deitel, Java: How to Program, 9th Edition (2011), Pearson College Div
2. Kathy Sierra and Bert Bates, Head First Java, 2nd Edition (2005), O'Reilly Media
3. Others